SMART PARKING SYSTEM BASED ON IOT

IoT-Based Smart Parking Management System Using RFID and Ultrasonic Sensors

Abstract

This paper presents an IoT-based smart parking management system using Arduino Uno, RFID, and ultrasonic sensors. The proposed system automatically verifies authorized vehicles through RFID authentication and monitors parking space availability using ultrasonic sensors. A servo motor is used to control the parking gate based on authentication results and parking availability. The system aims to improve parking management efficiency while maintaining low implementation cost and simple deployment.

Keywords:

IoT, Smart Parking, RFID Authentication, Ultrasonic Sensor, Arduino Uno, Parking Management

1. Problem Statement

Project Topic

Low-Cost IoT-Based Smart Parking System Using Arduino Uno and Ultrasonic Sensors

Problem Statement

Many parking lots still rely on manual management, making vehicle verification and parking space monitoring inefficient. An automated system is needed to authenticate vehicles and provide real-time parking availability information.

Objective

Develop an IoT-based smart parking system using RFID for vehicle authentication and ultrasonic sensors for parking space detection. The system aims to automatically verify vehicles and monitor available parking spaces in real time.

2. Search Strategy

The keywords used include **Smart Parking System**, **RFID Parking**, **IoT Parking Management**, **Parking Space Detection**. More than 20 papers were identified, and 6 papers were selected based on their relevance to RFID authentication, parking occupancy detection, and IoT implementation.

3. Literature Summary

Table 1: Literature Summary

Paper	Problem	Solution	Technology	Metrics	Limitation
P1 (Ultrasonic Control Sensors, 2021)	Time-consuming parking search and inefficient parking management.	Smart parking system using RFID authentication and ultrasonic-based slot detection.	Arduino Mega 2560, RFID, HC-SR04, LCD Display, Servo Motor, ESP8266.	Parking status monitoring; Reduced vehicle search time.	Limited prototype; Requires Wi-Fi infrastructure.
P2 (IoT Parking, 2019)	Heavy traffic jams and energy wastage in urban areas.	4-slot automated system with real-time web monitoring and gate control.	Arduino Mega, HC-SR04, Servo Motor, Ethernet Shield.	Real-time availability; Fuel savings.	Dependent on wired LAN connection.
P3 (Smart Car Parking Using IIoT, 2025)	Parking congestion and inefficient manual monitoring.	Smart parking system using RFID authentication and IoT-based monitoring.	ESP32, RFID, IR Sensors, LCD Display, Servo Motor.	Real-time monitoring; Faster vehicle authentication.	IR sensors are sensitive to environmental conditions.
P4 (Arduino and Ultrasonic, 2025)	Difficulty in identifying available parking spaces and inefficient manual parking management.	Automated parking system with ultrasonic-based slot detection and LCD status display.	Arduino Uno, HC-SR04, LCD Display, Buzzer.	Real-time parking status display; Improved parking management efficiency.	Limited to a small-scale prototype; No remote monitoring or IoT connectivity.
P5 (Automated Gate Access, 2025)	Mismanagement of parking areas and high latency in updates.	Distributed sensing with synchronized real-time gate control.	Arduino Uno, NodeMCU, HC-SR04, LCD Display, Servo Motor.	Success rate (¿95%); Response time (1–2 s).	Prototype scale lacks long-term outdoor testing.
P6 (Arduino and IR Sensors, 2023)	Manual parking management and inefficient vehicle guidance.	Smart parking system using IR sensors for vehicle detection and automatic parking status updates.	Arduino Uno, IR Sensors, LCD 16×2 (I2C), Servo Motor.	Automatic vehicle detection; Real-time parking status display.	IR sensors are sensitive to ambient light and obstacle interference.

4. Comparative Analysis

Table 2: Comparative Analysis

Paper	RFID	HC-SR04	LCD 1602A	I2C	ESP32	Arduino Uno	Servo Motor
P1	✓	✓	✓				✓
P2		✓					✓
P3	✓		✓		✓		✓
P4		✓	✓			✓	
P5		✓	✓			✓	✓
P6			✓	✓		✓	✓

Trade-off:

- **RFID:** Provides fast and reliable vehicle authentication at a low cost. However, it requires registered RFID cards and cannot identify unauthorized vehicles.
- **HC-SR04:** Low-cost and effective for vehicle detection over short distances. However, its performance may be affected by environmental conditions and has a limited sensing range.
- **LCD 1602A:** Provides clear real-time parking information at a low cost. However, it has limited display capacity and supports only simple text.
- **I2C:** Reduces wiring complexity and Arduino I/O pin usage. However, it requires an additional I2C module and supports fewer devices than parallel communication.
- **ESP32:** Provides built-in Wi-Fi connectivity and supports IoT applications. However, it consumes more power and is more complex to configure than Arduino Uno.
- **Arduino Uno:** Easy to program, inexpensive, and suitable for small-scale embedded projects. However, it has limited processing power and memory for complex applications.
- **Servo Motor:** Provides precise and reliable automatic gate control with easy Arduino integration. However, it is mainly suitable for lightweight prototype systems and has limited torque.

5. Research Gap

Gap 1

Most existing studies implement RFID-based vehicle authentication, but only a few evaluate RFID authentication response time in low-cost Arduino-based smart parking systems.

Gap 2

Although RFID authentication and parking occupancy detection have been widely studied separately, limited research evaluates the overall performance when both technologies are integrated into a single Arduino-based smart parking platform.

6. Research Questions

Based on the identified research gaps, this study proposes the following research questions.

1. How accurate is RFID authentication in an Arduino-based smart parking system?
2. How accurate are ultrasonic sensors in detecting parking slot occupancy in an Arduino-based smart parking system?

7. Design Decision

Table 3: Design Decision

Element	**Options to consider**	**Group's choice**	**Evidence from LR**	**Technical reasons**
MCU / Controller	Arduino Uno, ESP32, NodeMCU	Arduino Uno	P4, P5, P6	Easy to program, low cost, and suitable for hardware control.
RFID Reader	RFID RC522, RFID PN532, NFC Reader	RFID RC522	P1, P3	Low cost, reliable authentication, and easy Arduino integration.
Parking Detection Sensor	Ultrasonic HC-SR04, IR Sensor	Ultrasonic HC-SR04	P1, P2, P4, P5	Accurately detects vehicle entry and exit.
Gate Control	Servo Motor, DC Motor	Servo Motor SG90	P1, P2, P3, P5, P6	Precise angle control and easy integration.
Display	LCD 16x2, OLED Display, Seven-Segment Display	LCD 1602 16x2	P1, P3, P4, P5, P6	Displays parking information while reducing I/O pin usage.
IoT Monitoring	Blynk, ThingSpeak, Firebase, MQTT Dashboard	Blynk IoT Platform	P2, P3	Provides real-time remote monitoring.

8. Proposed Block Diagram

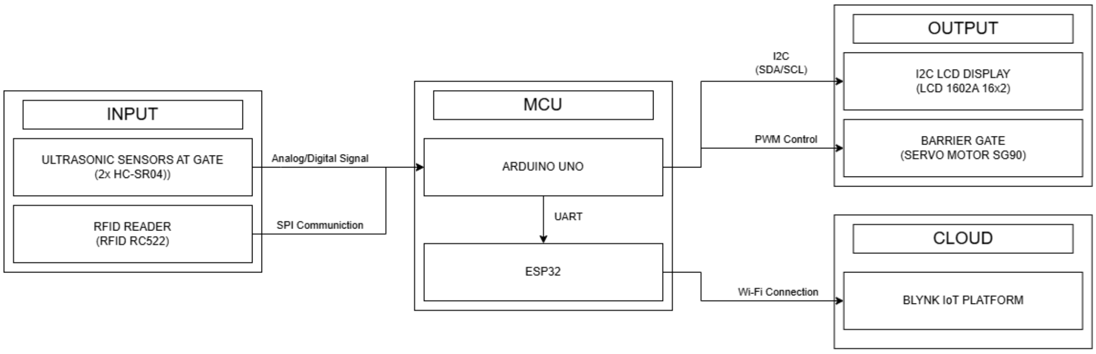

Figure 1: Proposed Smart Parking System Block Diagram

Table 4: Mapping Table from Literature Review

Elements from Block Diagram	**Evidence from LR**	**Reason**
RFID Card/Tap RFID Reader RC522	P1,P3	Low cost, fast authentication, and easy integration.
Ultrasonic Sensor HC-SR04	P1,P2,P4,P5	Find the perfect parking spot at a low cost.
Arduino Uno	P4,P5,P6	Easy to program and suitable for low-cost projects.
Servo Motor	P1,P2,P3,P5,P6	Control automatic gates accurately and simply.
LCD 1602A I2C	P1,P3,P4,P5,P6	Displays visual information in real time.
ESP32	P3	Provides Wi-Fi connectivity for IoT communication.
Cloud	P2,P3	Enables remote data storage and real-time monitoring.

9. Expected Metrics

Table 5: Expected Metrics

System Type	Metrics	Unit	Linked RQ
Smart Parking	Authentication Accuracy	%	RQ1
Smart Parking	Detection Accuracy	%	RQ2
Smart Parking	Authentication Response Time	ms	RQ1

10. AI Reflection

Table 6: AI Reflection

Prompt	AI Output	Verification	Reflection
Compare HC-SR04 and IR sensors for parking slot detection.	HC-SR04 is more reliable, while IR sensors are more affected by environmental factors.	Verified through reference materials that HC-SR04 is more suitable for vehicle detection.	Chose HC-SR04 as the main sensor for the system.
Should our Smart Parking project use Arduino Uno or ESP32 for IoT communication?	The AI recommended using ESP32 because it has built-in WiFi, more processing power, and more memory.	Compared Arduino Uno and ESP32 from their official documentation, which include wireless communication and an end-to-end IoT system.	Our team use both ESP32 and Arduino Uno for better scalability and wireless communication.

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